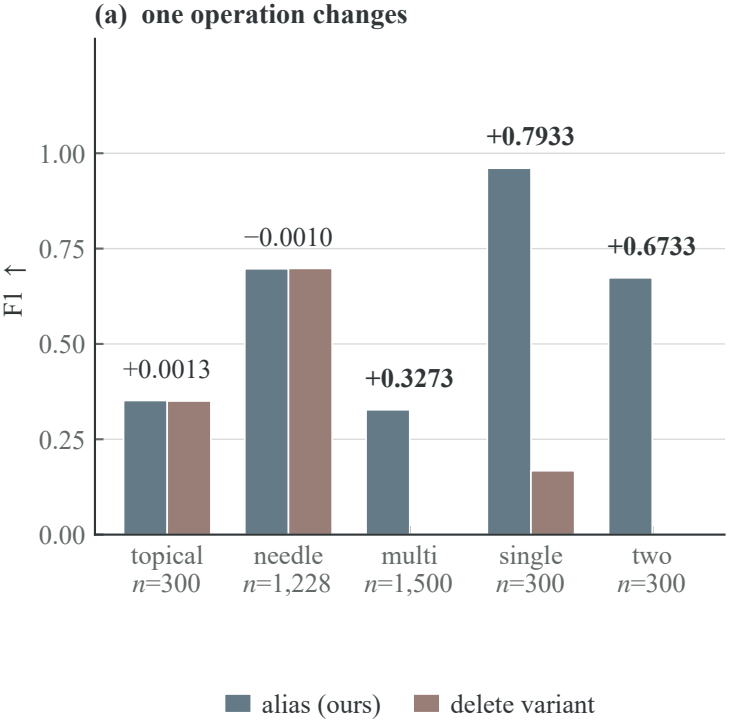
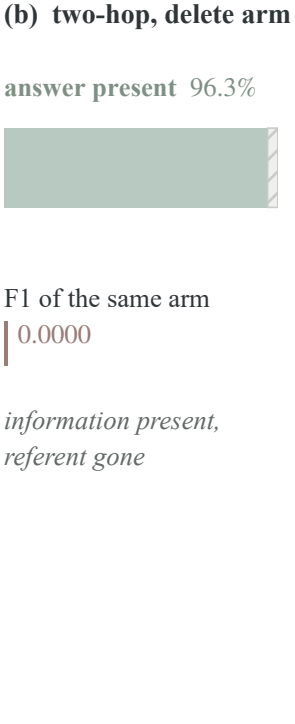

(a) one operation changes
F1 ↑
1.00
0.75
0.50
0.25
0.00
+0.0013
−0.0010
+0.3273
+0.7933
+0.6733
topical
n=300
needle
n=1,228
multi
n=1,500
single
n=300
two
n=300
alias (ours)
delete variant


(b) two-hop, delete arm
answer present 96.3%
F1 of the same arm
0.0000
information present,
referent gone